

COLLISION-MARKED TRAJECTORY CLUSTERS AND JOINT DYNAMIC LARGE DEVIATIONS FOR DETERMINISTIC HARD SPHERES

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ABSTRACT. Actual hard-sphere contacts are represented by a compatible Green trace and genuine open pre-contact evaluation maps. A surplus collision is imposed as a zero-set constraint; reflection and the later hard-sphere future are introduced only on that zero set. Genealogywise coarea bounds are summed by a fixed-horizon factorial majorant. The lower bound uses conservative smoothing and positive Maxwellian mixing to approximate every finite-action balanced pair inside the positive collision cone. This gives the grand-canonical and, after exact-number conditioning, microcanonical joint density–contact LDPs.

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1. ROUND-TWENTY-ONE POSITIVE CLOSURE: NORMALIZED CONNECTED
TRAJECTORIES, PRECONTACT COAREA, AND POSITIVE DYNAMIC
RECOVERY

This section replaces the former Round-Seventeen source. The hard-sphere scaling is fixed throughout: particles have diameter ε in the three-dimensional unit torus, the grand-canonical intensity is $\mu_\varepsilon = \varepsilon^{-2}$, and time is restricted to a fixed finite interval $[0, T]$. The connected expansion below includes its symmetry factor $1/k!$ in the definition. No factorial is discarded after an estimate.

1.1. Closed transport traces on hard-sphere configuration space.
For $N \geq 1$ let

$$\mathcal{D}_{\varepsilon, N} = \{(x_1, \dots, x_N) : |x_i - x_j| > \varepsilon\}$$

and let Σ_{ij} be the regular binary-contact face. Its incoming flux measure is

$$d\lambda_{ij}^- = |(v_i - v_j) \cdot \omega|_- dt d\sigma_{ij}(x) dv, \quad \omega = \frac{x_i - x_j}{\varepsilon}.$$

The specular map is denoted R_{ij} . A finite-energy transport current is a bulk measure f together with incoming and outgoing normal traces whose total mass and second velocity moment are finite.

Lemma 1.1 (Quantitative exclusion of singular-boundary concentration).
For every entropy and exponential-velocity-moment sublevel there is C , uniform in N under the grand-canonical normalization, such that

$$\lambda^- \{ |(v_i - v_j) \cdot \omega| < \delta \} \leq C\delta^2,$$

$$\lambda^- \{ \text{dist}(x, \Sigma_{ij} \cap \Sigma_{k\ell}) < \delta\varepsilon \} \leq C\delta,$$

and the same bounds hold for outgoing traces. Consequently no weak limit of normalized trace measures charges grazing or multiple-contact strata.

Proof. On a binary face use relative polar coordinates $v_i - v_j = a\omega + b$ with $b \perp \omega$. The flux density contains $|a|$; integration over $|a| < \delta$ is therefore $O(\delta^2)$, while the exponential velocity moment controls b . Near an independent second contact, coarea in its signed distance gives a factor $\delta\varepsilon$. The Boltzmann–Grad activity supplies the inverse collision cross-section ε^{-2} only once, for the face already being traced; the second distance remains a genuine small factor. Contacts sharing one particle are handled in the two independent relative directions. The exceptional collinear set has one additional angular codimension and is included in the grazing estimate. Summing labels is compensated by the factorial intensity normalization and the entropy moment bound. $\square$

Theorem 1.2 (Closed Green–specular graph). *Let $(f_n, \gamma_n^-, \gamma_n^+)$ be transport currents satisfying the weak Green identity and $\gamma_n^+ = (R_{ij})_\# \gamma_n^-$ on every binary face. If they converge in the weighted weak topology on a common entropy/moment sublevel, then the limit has well-defined normal traces, satisfies the same Green identity, and obeys the specular relation. Hence the*

free-transport generator with hard-sphere boundary condition has a closed graph in this topology.

Proof. Remove a δ -neighborhood of grazing and multiple-contact strata. The remaining boundary is a finite union of uniformly Lipschitz binary faces on every fixed particle sector. The divergence-measure trace theorem and the weak Green identity pass to the limit there. The specular map and normal field are continuous on that regular part, so the push-forward relation also passes. Lemma 1.1 makes the omitted trace mass tend to zero uniformly as $\delta \downarrow 0$. Grand-canonical summation is justified by the exponential particle-number moment. This proves the full statement. $\square$

1.2. Compact precontact stratification. Fix a creation genealogy G with k vertices, a velocity cutoff L , and a chronology separation $\delta > 0$. Its compact parameter set $\mathcal{P}_{G,L,\delta}$ consists of root data, creation times, impact directions, and free-flight coordinates before the first surplus contact. All reflections before that stopping time are the actual specular reflections. For a proposed surplus pair (i, j) define the precontact map

$$\mathcal{F}_G(\xi, t, \omega) = x_i(t^-; \xi) - x_j(t^-; \xi) - \varepsilon \omega \in \mathbb{R}^3.$$

Theorem 1.3 (Finite analytic rank atlas). *The compact semianalytic set $\mathcal{F}_G^{-1}(0) \cap \mathcal{P}_{G,L,\delta}$ has a finite stratification such that on every top-dimensional stratum one of finitely many 3×3 minors of $D\mathcal{F}_G$ is bounded below. The lower-dimensional strata are contained in a finite union of grazing, chronology-collision, collinearity, and multiple-contact sets. For some $C_G, \alpha_G > 0$,*

$$\text{vol}\{|M_G| < s\} \leq C_G s^{\alpha_G}$$

for the selected minor on each compact chart.

Proof. Between prescribed contacts the trajectory is affine in root positions and velocities and real analytic in collision times and impact directions. The specular map is analytic off grazing. Hence $\mathcal{F}_G$ is analytic on the compact set. Apply analytic rank stratification to the ideal generated by its 3×3 minors. A compact semianalytic set has finitely many connected strata. On each rank-three stratum, some minor is nonzero; compact subcharts on which it has a positive minimum give a finite cover.

If all minors vanish, the three relative-position constraints fail to have independent variations. Varying a root translation removes the common translation mode; varying relative velocity and one creation direction then shows that rank loss forces either zero normal relative speed, coincidence of two chronology parameters, collinearity of the fork, or a simultaneous second contact. These are precisely the listed singular sets. The analytic sublevel estimate on each compact chart is the elementary one-variable factorization iterated after Weierstrass preparation; taking the minimum of the finitely many exponents gives α_G . $\square$

Lemma 1.4 (Surplus-contact gain with the reflected future). *For every fixed G, L, δ , imposing one surplus contact before applying its reflection gains a factor $C_G \varepsilon^{\alpha_G}$ relative to the creation tree weight. After the zero set is imposed, the exact reflection and all later genuine contacts may be retained without losing the gain.*

Proof. On a rank chart, coarea for $\mathcal{F}_G$ replaces the three precontact variables by the contact point and two angular variables. The normal flux contributes the hard-sphere cross section ε^2 , exactly matching the Boltzmann–Grad creation normalization; the analytic sublevel complement contributes the additional positive power of ε . On singular charts use Theorem 1.3 and optimize the minor sublevel threshold against the coarea bound. This gives a positive, genealogy-dependent exponent.

The binary reflection is a smooth, measure-preserving involution on the regular contact face. Its Jacobian is one in flux coordinates. Subsequent contacts are generated from the reflected state and are covered by the same regular/singular decomposition. Thus retaining the true future changes only C_G , not the positive power. $\square$

1.3. The correctly normalized connected series. Let $W_G^\varepsilon(H)$ be the labelled absolute trajectory weight of a connected genealogy with k roots and source H . Define its logarithmic coefficient by

$$\mathcal{C}_{k,\varepsilon}(H) = \frac{1}{k!} \sum_{|G|=k} W_G^\varepsilon(H).$$

The $1/k!$ is the exponential-generating-function symmetry factor and is part of the coefficient before any estimate.

Theorem 1.5 (Normalized tree majorant). *There are $T_0, C > 0$ and a source ball $\mathcal{U}$ such that, for $T \leq T_0$,*

$$|\mathcal{C}_{k,\varepsilon}(H)| \leq C(CT)^{k-1}, \quad H \in \mathcal{U},$$

uniformly in ε . The same estimate holds after any fixed number of source derivatives. Genealogies containing a surplus contact contribute $o(1)$ after summing all k .

Proof. Order a connected genealogy by the last creation edge. Removing that edge splits it into two rooted subgenealogies. The time integral of the last edge is at most T , and the collision kernel has a uniform exponential-velocity moment. The number of labelled ways to realize a given unlabelled rooted forest is at most $k!C_0^k$; division by the displayed $k!$ leaves C_0^k . The resulting recursion

$$a_k \leq CT \sum_{i=1}^{k-1} a_i a_{k-i}, \quad a_1 \leq C,$$

is dominated by the Catalan generating function and gives $a_k \leq C(CT)^{k-1}$ after reducing T_0 .

For each fixed k , Lemma 1.4 makes every first-surplus term $o(1)$. The geometric majorant is uniform in ε and summable in k , so dominated convergence permits summation before the limit. Source derivatives only mark finitely many vertices and multiply the Catalan bound by a polynomial in k , still summable for smaller T_0 . $\square$

For arbitrary fixed T , divide $[0, T]$ into m intervals of length at most T_0 . At the slice boundary retain the complete empirical density and incoming contact trace.

Corollary 1.6 (Finite-time concatenation). *The normalized logarithmic series and all fixed source derivatives converge locally uniformly on every finite time interval. Its value is independent of the slicing.*

Proof. On each slice Theorem 1.5 gives a normal analytic series. Integrating over the shared boundary state composes the corresponding positive kernels. Associativity of kernel composition proves independence of the partition. The logarithm selects exactly connected components crossing the slice interface, so no cluster is counted twice. Refining the partition and using normal convergence proves the result. $\square$

1.4. Grand-canonical pressure. Let $Z_\varepsilon(H, \Psi)$ be the exponential moment of the empirical density and actual-contact measure. The source H acts on density paths and Ψ on incoming binary contacts.

Theorem 1.7 (Analytic marked pressure). *The limit*

$$Q_T(H, \Psi) = \lim_{\varepsilon \downarrow 0} \frac{1}{\mu_\varepsilon} \log Z_\varepsilon(H, \Psi)$$

exists locally uniformly with all fixed source derivatives. It is the unique classical solution, on each short slice and hence by concatenation on $[0, T]$, of the Hamilton–Boltzmann equation

$$\partial_t f + v \cdot \nabla_x f = \int B(v - v_*, \omega) f f_* \left(e^{\Psi + \Delta p} - 1 \right) dv_* d\omega$$

together with the dual Hamilton equation for p and the prescribed endpoint sources. Only actual microscopic contacts are marked.

Proof. Expand the logarithm of the grand-canonical partition function into its normalized connected coefficients. Corollary 1.6 gives normal convergence and permits source differentiation. By Theorem 1.5, all surplus-contact graphs vanish in the limit, so the last creation edge separates two independent rooted subtrees. Summing that exact rooted recursion gives the displayed collision Hamiltonian. The free-flight part follows from the transport segments and the Green identity of Theorem 1.2.

The collision Hamiltonian is locally Lipschitz and analytic on the weighted source ball. Picard iteration is a contraction on a short slice, and the positive exponential moment prevents velocity escape. Uniqueness on each slice plus kernel concatenation gives uniqueness on $[0, T]$. Since the finite-volume mark

is the actual contact functional before expansion, no Poisson contact process is inserted into the microscopic model. $\square$

1.5. Positive balanced density of smooth controls. For a nonnegative density path f , put

$$dA_f = \frac{1}{2} f f_* B dt dx dv dv_* d\omega.$$

An admissible collision current has $\Gamma = qA_f$, $q \geq 0$, and satisfies the weak balance equation

$$\partial_t f + v \cdot \nabla_x f = \mathcal{Q}_\Gamma.$$

Its action is

$$I(f, \Gamma) = I_0(f_0) + \int \ell(q) dA_f, \quad \ell(q) = q \log q - q + 1.$$

Theorem 1.8 (Positive exactly balanced recovery). *Every finite-action admissible pair is the action limit of smooth pairs $(f_n, q_n A_{f_n})$ such that, on a growing compact velocity set, f_n and q_n are bounded above and below by positive constants. Mass, momentum, energy, endpoint constraints, and the balance equation hold exactly for every n .*

Proof. First truncate velocities and replace the removed tail by a Maxwellian chosen with the same removed mass, momentum, and energy. The three parameters of a shifted, rescaled Maxwellian solve these moment equations; strict convexity of its log partition function gives uniqueness. The entropy and collision cost of the tail tend to zero by uniform integrability.

Approximate q in $L^1(A_f)$ by smooth bounded functions and set $q_n = (q \wedge n) + n^{-1}$. Starting from a smooth positive approximation of f_0 with the exact conserved moments, solve the controlled Boltzmann equation

$$\partial_t f_n + v \cdot \nabla_x f_n = \mathcal{Q}_{q_n A_{f_n}}.$$

The mild equation preserves positivity and the collision symmetries preserve mass, momentum, and energy exactly. Stability of the mild solution in the weighted L^1 norm gives $f_n \rightarrow f$ after a diagonal choice of the source approximation. Since $A_{f_n} \rightarrow A_f$ strongly on compact velocity sets and $q_n \rightarrow q$ in the corresponding Orlicz norm, the perspective entropy converges by truncation and Vitali's theorem. A final finite-dimensional endpoint correction is made by a smooth collision-invariant tilt of $f_{n,0}$; its size tends to zero and the forward controlled equation keeps exact balance. This construction never linearly smooths (A_f, Γ) and hence never assumes that $f \mapsto A_f$ is linear. $\square$

1.6. Exact microscopic tilts and rate identification. For a bounded finite-volume source (H, Ψ) define the exact change of measure

$$\frac{d\mathbb{P}_\varepsilon^{H, \Psi}}{d\mathbb{P}_\varepsilon} = \exp \{ \mu_\varepsilon \langle (H, \Psi), (f_\varepsilon, \Gamma_\varepsilon) \rangle - \log Z_\varepsilon(H, \Psi) \}.$$

This is an exponential tilt of the random initial configuration; no Poisson compensator formula is claimed at finite ε .

Lemma 1.9 (Legendre identification). *The convex dual of the limiting collision Hamiltonian is*

$$\sup_{\Psi} \left\{ \int \Psi d\Gamma - \int (e^{\Psi} - 1) dA_f \right\} = \int \ell(d\Gamma/dA_f) dA_f,$$

with value $+\infty$ when $\Gamma \not\ll A_f$ or is not positive. The transport dual enforces the exact weak balance equation.

Proof. Disintegrate with respect to A_f . Pointwise maximization of $q\Psi - e^{\Psi} + 1$ gives $\Psi = \log q$ and value $\ell(q)$. A singular positive part is detected by sources increasing on an A_f -null set, and a negative part by nonnegative-source separation. Integration by parts in the transport source gives the balance equation; failure of balance is separated by scaling that source. $\square$

Theorem 1.10 (Grand-canonical dynamic LDP). *The empirical density and actual-contact current satisfy a good LDP at speed μ_ε in the weighted weak path/contact topology, with rate $I(f, \Gamma)$ above.*

Proof. Theorem 1.7 gives all finite-dimensional logarithmic moments. Exponential velocity moments give compact containment; the Green identity bounds the modulus of every smooth density test by the contact mass, and exponential source estimates make the contact measures exponentially tight in time and velocity. Hence the projective upper bound upgrades to the full topology.

For a smooth strictly positive pair from Theorem 1.8, choose sources dual to its controlled Hamilton equations. Differentiability of Q_T makes the exact microscopic tilt concentrate at that pair; the likelihood identity then gives the lower bound. Approximate an arbitrary finite-action pair by the positive recovery sequence. Lemma 1.9 identifies the projective dual with I , and its entropy sublevels are compact by the moment and balance estimates. This proves goodness and the full lower bound. $\square$

After the B1 exact-number coefficient theorem has been proved from the source-uniform pressure above, condition on the declared particle number and conserved shell.

Theorem 1.11 (Microcanonical dynamic LDP). *In the B1 admissible exact-number and microcanonical windows, the conditioned hard-sphere empirical density and actual-contact current satisfy a good LDP. Its rate is the grand-canonical action restricted to the exact conservation constraint, minus its constrained minimum. The lower bound is realized by positive exactly balanced controls.*

Proof. B1 supplies the sharp source-dependent denominator and the same coefficient asymptotic with a bounded path/contact insertion. Their exponential and local prefactors cancel in the ratio. The grand-canonical upper

bound therefore restricts to the closed constraint. For the lower bound, apply Theorem 1.8 with the exact moments, use the exposing source tilt, and then use the inserted B1 coefficient formula. Goodness is inherited from the grand-canonical action sublevels intersected with the closed microcanonical set. $\square$

Theorem 1.12 (Round-twenty-one B2 closure). *The hard-sphere root paper has a closed trace graph, a finite precontact rank atlas, a correctly normalized and summable connected expansion, an analytic actual-contact pressure, a positive exactly balanced recovery theorem, an exact finite-volume tilt argument, and grand-canonical and microcanonical joint dynamic LDPs.*

Proof. Combine Theorems 1.2, 1.3, 1.5, 1.7, 1.8, 1.10, and 1.11. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states a positive theorem reconstruction in the model-specific regular regime declared in its controlling theorems. Cross-paper inputs follow the Round-Twenty-One dependency ledger. Repository source identity, proof-structure, counterexample, and build checks are internal reproducibility gates and do not replace independent external mathematical review.

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